⚡ EXAM-READY FORMULA SHEET

Crystallography & X-ray Diffraction

Unit II — Engineering Physics

SUBJECT **Engineering Physics**	UNIT **Unit II**
LEVEL **B.Tech 1st Year**	SEMESTER **1st Semester**

📌 Formulas 📊 Tables 📇 Flashcards ⚡ Exam Tips 🚀 Quick Revision

This document contains all key formulas, concepts, and exam strategies for Unit II

📌 COURSE OBJECTIVES

OBJ 1
Understand crystal structures and periodic lattice arrangements

OBJ 2
Identify all 7 crystal systems and 14 Bravais lattices

OBJ 3
Apply Miller indices and calculate interplanar spacing

OBJ 4
Understand X-ray diffraction and Bragg's law derivation

OBJ 5
Determine crystal structure using Laue and Powder methods

OBJ 6
Calculate APF, coordination number and packing efficiency

🔷 CRYSTALLOGRAPHY BASICS — KEY DEFINITIONS

Space Lattice	A regular, periodic, infinite 3D arrangement of points in space; each point has identical surroundings.
Basis (Motif)	The atom or group of atoms associated with each lattice point; defines what is placed at every lattice point.
Crystal Structure	The actual physical structure formed by placing the basis on every lattice point of the space lattice.
Lattice Parameter	The edge lengths (a, b, c) and inter-axial angles (α, β, γ) that define a unit cell.
Primitive Lattice	Lattice with atoms only at the corners; contains effectively ONE atom per unit cell.
Non-primitive Lattice	Contains atoms at corners plus face/body/edge centers; more than one atom per unit cell.

Crystal Structure = Space Lattice + Basis

The crystal is generated by repeating the basis at every point of the space lattice

📦 UNIT CELL & LATTICE PARAMETERS

The **smallest repeating unit** that, when translated in 3D space, generates the entire crystal lattice.

Edge lengths	a, b, c (Ångström or nm)
Angle α	Between b and c axes
Angle β	Between a and c axes
Angle γ	Between a and b axes

ATOMS PER UNIT CELL

Corner atom → 1/8 contribution
Face atom → 1/2 contribution
Edge atom → 1/4 contribution
Body atom → 1 (full)

SC | BCC | FCC — ATOMS PER CELL

SC = 8 × (1/8) = 1
BCC = 1 + 8×(1/8) = 2
FCC = 6×(1/2) + 8×(1/8) = 4

🧱 7 CRYSTAL SYSTEMS + 14 BRAVAIS LATTICES

Complete Table

#	Crystal System	Axis Relations	Angle Conditions	Bravais Lattices	Lattice Types	Example
1	**Cubic**	$a = b = c$	$\alpha = \beta = \gamma = 90°$	3	P, I, F	NaCl, Cu, Fe
2	**Tetragonal**	$a = b \neq c$	$\alpha = \beta = \gamma = 90°$	2	P, I	TiO_2, Sn
3	**Orthorhombic**	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90°$	4	P, I, F, C	$BaSO_4$, S
4	**Monoclinic**	$a \neq b \neq c$	$\alpha = \gamma = 90° \neq \beta$	2	P, C	$CaSO_4$, Gypsum
5	**Triclinic**	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90°$	1	P	$K_2Cr_2O_7$
6	**Hexagonal**	$a = b \neq c$	$\alpha = \beta = 90°,\ \gamma = 120°$	1	P	Zn, Mg, Graphite
7	**Rhombohedral** (Trigonal)	$a = b = c$	$\alpha = \beta = \gamma \neq 90°$	1	P	Calcite, As, Sb

📎 *Lattice types: P = Primitive | I = Body-centered | F = Face-centered | C = Base-centered | Total = 14*

🔢 COORDINATION NUMBER & ATOMIC PACKING FACTOR

ATOMIC PACKING FACTOR (APF) DEFINITION

APF = (Number of atoms/unit cell × Volume of one atom) / Volume of unit cell

VOLUME OF ONE ATOM (SPHERE)

$$V_{atom} = (4/3)\ \pi\ r^3$$

Structure	Atoms/Cell	Coord. No.	Relation (r & a)	APF Formula	APF Value	Examples
SC (Simple Cubic)	1	6	$a = 2r$	$\pi/6$	≈ 0.52 (52%)	Po (only)
BCC (Body-Centered)	2	8	$4r = a\sqrt{3}$	$\sqrt{3}\pi/8$	≈ 0.68 (68%)	Fe(α), W, Cr, Na
FCC (Face-Centered)	4	12	$4r = a\sqrt{2}$	$\pi/(3\sqrt{2})$	≈ 0.74 (74%)	Cu, Al, Ag, Au

⭐ *FCC has the highest packing efficiency (74%) — most closely packed structure*

⭐ *Coordination Number = number of nearest neighbours of an atom*

MILLER INDICES — Definition, Steps & Properties

DEFINITION

A set of three integers (h, k, l) that describe the orientation of a crystal plane in terms of its intercepts on the crystallographic axes. Denoted as **(hkl)**.

Steps to Find Miller Indices

1. Identify where the plane intercepts the x, y, z axes. Express as multiples of unit cell lengths: e.g., (2a, 3b, 1c)
2. Take the **reciprocals** of intercepts: 1/2, 1/3, 1/1
3. **Clear fractions** by multiplying by LCM: ×6 → 3, 2, 6
4. Write as **(hkl)** = (3 2 6) — no commas
5. Negative intercepts: write bar over index: e.g., (-1) → $\bar{1}$

MILLER INDEX NOTATION

(hkl) → Single plane or family of parallel planes
{hkl} → All equivalent planes (family)
[uvw] → A specific direction
<uvw> → All equivalent directions

Important Properties of Miller Indices

- Parallel planes have the same Miller indices
- A plane parallel to an axis has index 0 for that axis
- Plane (hkl) is perpendicular to direction [hkl] in cubic systems
- If plane cuts axis on negative side, bar notation is used: ($\bar{h}$kl)
- In cubic system: $d_{(hkl)} = a / \sqrt{h^2+k^2+l^2}$

Plane	Description	Inter-spacing
(100)	Perpendicular to x-axis	$d = a$
(110)	Diagonal plane	$d = a/\sqrt{2}$
(111)	Body diagonal plane	$d = a/\sqrt{3}$
(200)	Parallel to (100)	$d = a/2$

INTERPLANAR SPACING FORMULA

INTERPLANAR SPACING — CUBIC SYSTEM (MOST IMPORTANT)

$$d_{(hkl)} = a / \sqrt{h^2 + k^2 + l^2}$$

$d_{(hkl)}$	Perpendicular distance between adjacent parallel (hkl) planes
a	Lattice parameter (edge length of unit cell)
h, k, l	Miller indices of the plane

⚠️ *This formula is valid ONLY for the cubic system*

GENERAL FORMULA (ORTHORHOMBIC)

$$1/d^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$$

TETRAGONAL SYSTEM

$$1/d^2 = (h^2+k^2)/a^2 + l^2/c^2$$

X-RAY DIFFRACTION — BASICS

WHAT ARE X-RAYS?

Electromagnetic radiation with wavelength 0.01–10 Å. Comparable to interatomic spacing in crystals, enabling diffraction.

WHY X-RAY DIFFRACTION?

Crystals act as 3D diffraction gratings. Bragg planes reflect X-rays; constructive interference reveals crystal structure.

DIFFRACTION CONDITION

Constructive interference occurs when path difference = integer multiple of wavelength → peaks in diffraction pattern.

BRAGG'S LAW — The Most Important Formula

$$n\lambda = 2d\sin\theta$$

Condition for constructive X-ray diffraction from crystal planes

n	Order of diffraction (integer: 1, 2, 3,...); usually n = 1 for first order
λ	Wavelength of incident X-rays (Å or nm)
d	Interplanar spacing $d_{(hkl)}$ (Å or nm)
θ	Glancing angle (angle between incident beam and crystal plane — NOT the normal)

DERIVED FROM BRAGG'S LAW

$\sin\theta = n\lambda / 2d$

$d = n\lambda / (2\sin\theta)$

$\lambda = 2d\sin\theta / n$ (when n=1)

⚠️ **Critical: θ is GLANCING angle**

- θ is measured from the plane surface, NOT the normal to the plane
- Diffraction angle 2θ is what the detector measures
- Maximum diffraction at 2θ = 180° → $\sin\theta_{max} = 1$ → $d_{min} = \lambda/2$

🔑 Bragg's Law — Physical Reasoning

1. X-rays strike crystal planes at glancing angle θ. Rays reflected from adjacent planes travel different path lengths.
2. Path difference between rays from adjacent planes = 2d sinθ (geometric derivation using triangle formed by incident and reflected rays).
3. For constructive interference: Path difference = nλ → Therefore: **nλ = 2d sinθ**

X-RAY DIFFRACTOMETER (XRD)

X-RAY SOURCE

Hot cathode tube emits characteristic X-rays (Cu Kα: λ=1.54 Å most common). Monochromator filters to single wavelength.

SPECIMEN

Powdered crystalline sample placed on rotating stage. Different crystallite orientations satisfy Bragg's law at different θ.

DETECTOR

Geiger counter or scintillation detector rotates at 2θ for every θ rotation of sample. Records intensity vs 2θ angle.

STRUCTURE DETERMINATION METHODS — COMPARISON

Property	Laue Method	Rotating Crystal Method	Powder (Debye-Scherrer) Method
X-ray Type	Polychromatic (white X-rays, broad λ)	Monochromatic (single λ)	Monochromatic (single λ)
Sample Type	Single crystal (stationary)	Single crystal (rotated)	Polycrystalline powder
Variable	λ (wavelength varies); θ is fixed	θ (angle changes as crystal rotates); λ fixed	θ (many orientations present); λ fixed
Crystal Orientation	Fixed — doesn't move	Rotated about one axis	Random (all orientations present in powder)
Record Medium	Flat photographic film	Cylindrical film	Cylindrical film or detector
Pattern	Spots (Laue spots)	Layer lines of spots	Concentric rings (Debye rings)
Main Use	Determining crystal symmetry and orientation	Determining lattice parameters	Identifying unknown crystalline materials; phase analysis
Advantage	Simple, fast; gives symmetry info quickly	Good for unit cell determination	No single crystal needed; most widely used
Limitation	Overlapping spots; not ideal for structure factors	Tedious; requires good single crystal	Overlapping peaks for complex structures

ALL IMPORTANT FORMULAS — SUMMARY TABLE

Formula Name	Expression	Variables	Used For
Bragg's Law	$n\lambda = 2d \sin\theta$	n=order, λ=wavelength, d=spacing, θ=glancing angle	XRD peak positions
Interplanar Spacing (Cubic)	$d = a / \sqrt{h^2+k^2+l^2}$	a=lattice parameter, h,k,l=Miller indices	Finding plane spacing
APF (SC)	$\pi/6 \approx 0.5236$	r=a/2 for SC	Packing efficiency
APF (BCC)	$\sqrt{3}\pi/8 \approx 0.6802$	$r=a\sqrt{3}/4$ for BCC	Packing efficiency
APF (FCC)	$\pi/(3\sqrt{2}) \approx 0.7405$	$r=a\sqrt{2}/4$ for FCC	Packing efficiency
Atoms/Cell (SC)	$N = 8 \times (1/8) = 1$	Corner contributions	Unit cell analysis
Atoms/Cell (BCC)	$N = 8\times(1/8) + 1 = 2$	Corner + body center	Unit cell analysis
Atoms/Cell (FCC)	$N = 8\times(1/8) + 6\times(1/2) = 4$	Corner + face centers	Unit cell analysis
SC radius relation	$a = 2r$	a=lattice param, r=atomic radius	Geometry of SC
BCC radius relation	$4r = a\sqrt{3}$	Body diagonal = 4r	Geometry of BCC
FCC radius relation	$4r = a\sqrt{2}$	Face diagonal = 4r	Geometry of FCC

FLASHCARDS — 25 Key Q&A for Exam

25 Cards

Q: What is a Space Lattice?

Regular periodic 3D arrangement of points where each point has identical surroundings

Q: What is a Basis (Motif)?

Atom or group of atoms attached to each lattice point

Q: Crystal Structure formula?

Crystal Structure = Lattice + Basis

Q: Total number of Bravais Lattices?

14 Bravais Lattices in 7 crystal systems

Q: Coordination number of SC?

6

Q: Coordination number of BCC?

8

Q: Coordination number of FCC?

12

Q: Atoms per unit cell in FCC?

4 atoms per unit cell

Q: APF of FCC?

$\pi/(3\sqrt{2}) \approx 0.74$ (74%)

Q: APF of BCC?

$\sqrt{3}\pi/8 \approx 0.68$ (68%)

Q: APF of SC?

$\pi/6 \approx 0.52$ (52%)

Q: Bragg's Law?

$n\lambda = 2d \sin\theta$

Q: What does θ represent in Bragg's Law?

Glancing angle — angle between X-ray beam and crystal plane surface (NOT normal)

Q: Interplanar spacing formula (Cubic)?

$d = a / \sqrt{(h^2 + k^2 + l^2)}$

Q: Cubic system conditions?

$a = b = c$; $\alpha = \beta = \gamma = 90°$

Q: Hexagonal system angles?

$\alpha = \beta = 90°$, $\gamma = 120°$; $a = b \neq c$

Q: Which crystal system has only 1 Bravais lattice?

Triclinic, Hexagonal, Rhombohedral (each has 1)

Q: Which crystal system has 4 Bravais lattices?

Orthorhombic (P, I, F, C)

Q: What method uses polychromatic X-rays?

Laue Method

Q: Which method uses polycrystalline powder?

Powder (Debye-Scherrer) Method

Q: Miller indices of plane parallel to YZ plane?

(100) — intercepts: 1, ∞, ∞ → reciprocal: 1, 0, 0

Q: What does {hkl} (curly brackets) denote?

Family of all equivalent planes related by symmetry

Q: Relation between d(100), d(110), d(111) for cubic?

$d(100):d(110):d(111) = 1 : 1/\sqrt{2} : 1/\sqrt{3}$

Q: What is the Debye-Scherrer equation used for?

Estimating crystallite size from peak broadening: $D = K\lambda/(\beta \cos\theta)$

Q: Minimum detectable d-spacing by X-ray?

$d_{min} = \lambda/2$ (when $\sin\theta = 1$, $\theta = 90°$)

EXAM TIPS — AVOID THESE COMMON MISTAKES

Mistake 1 — Miller Indices Steps

- Don't forget to take RECIPROCALS first, then clear fractions
- A plane parallel to an axis has intercept ∞ → Miller index = 0, not ∞
- Never write commas between indices: write (231) not (2,3,1)
- Negative index written as $\bar{1}$, not -1 or (-1)
- Always reduce to smallest integers: (246) → write as (123)

Mistake 2 — Bragg's Law Angle

- θ is the GLANCING angle (from plane), NOT the angle from normal
- The detector angle is 2θ (twice the glancing angle)
- $\sin\theta$ must be between 0 and 1; check your answer!
- If given $2\theta = 30°$, then $\theta = 15°$ for use in formula
- Higher order n means higher θ for same planes

Mistake 3 — Interplanar Spacing

- $d = a/\sqrt{h^2+k^2+l^2}$ is ONLY for cubic systems
- Always check units: a and d must be in same units (Å or nm)
- Larger h,k,l values → smaller d spacing → higher angle diffraction
- d(200) = d(100)/2 because h doubles

Mistake 4 — Crystal System Identification

- Cubic: ALL three sides equal AND all angles 90° — both conditions required
- Rhombohedral: all sides equal but angles ≠ 90° (often confused with cubic)
- Hexagonal has $\gamma = 120°$, not all angles equal
- Tetragonal: $a = b \neq c$ (one axis different, all angles 90°)
- Triclinic is the most GENERAL (no constraints equal)

Mistake 5 — Mixing SC, BCC, FCC

- SC: 1 atom/cell | BCC: 2 atoms/cell | FCC: 4 atoms/cell
- SC has 6 nearest neighbors (not 4 or 8)
- BCC body diagonal connects nearest neighbors ($\sqrt{3}$ a)
- FCC face diagonal connects nearest neighbors ($\sqrt{2}$ a)
- FCC ≠ HCP (both CN=12, but different stacking ABCABC vs ABABAB)

Mistake 6 — X-ray Methods

- Laue method = single crystal + polychromatic X-rays
- Powder method = polycrystalline + monochromatic X-rays
- Powder pattern shows RINGS not spots
- Laue pattern shows SPOTS arranged by symmetry

MEMORY TRICKS & MNEMONICS

APF ORDER

S < B < F
(SC=52% < BCC=68% < FCC=74%)
"Simple is Simplest, Face is Fullest"

CN MEMORY

6 → 8 → 12
SC=6, BCC=8, FCC=12
"6×8=48, 48×12 remember in order"

BRAGG'S LAW

$n\lambda = 2d\sin\theta$
"n Lambs Eat 2 Dogs Sitting"
$n \cdot \lambda = 2 \cdot d \cdot \sin\theta$

MILLER INDEX STEPS

I – R – C – E
Intercepts → Reciprocals → Clear fractions → Enclose in ()

CRYSTAL SYSTEMS COUNT

C-T-O-M-T-H-R
Cubic-Tetragonal-Ortho-Mono-Tri-Hexa-Rhombo
"Can Tigers Often Make Three Happy Roars?"

14 BRAVAIS

3+2+4+2+1+1+1 = 14
Cubic(3)+Tet(2)+Ortho(4)+Mono(2)+Tri(1)+Hex(1)+Rhombo(1)

⚡ QUICK REVISION SHEET

All formulas at a glance — No explanations — Pure expressions only

CORE FORMULAS

$n\lambda = 2d \sin\theta$	Bragg's Law
$d = a / \sqrt{h^2+k^2+l^2}$	d-spacing (Cubic)
$\sin\theta = n\lambda / 2d$	Bragg rearranged
$1/d^2 = h^2/a^2 + k^2/b^2 + l^2/c^2$	d-spacing (Orthorhombic)

PACKING FRACTIONS

$APF(SC) = \pi/6 \approx 0.52$	Simple Cubic
$APF(BCC) = \sqrt{3}\pi/8 \approx 0.68$	Body-Centered
$APF(FCC) = \pi/(3\sqrt{2}) \approx 0.74$	Face-Centered
$APF = N\cdot(4/3)\pi r^3 / a^3$	General APF

STRUCTURAL PARAMETERS

SC: $N=1$, $CN=6$, $a=2r$	SC values
BCC: $N=2$, $CN=8$, $4r=a\sqrt{3}$	BCC values
FCC: $N=4$, $CN=12$, $4r=a\sqrt{2}$	FCC values
$D = K\lambda/(\beta \cos\theta)$	Scherrer equation

CRYSTAL SYSTEMS

Cubic: $a=b=c$, $\alpha=\beta=\gamma=90°$	3 Bravais
Tetragonal: $a=b\neq c$, all $90°$	2 Bravais
Hexagonal: $a=b\neq c$, $\gamma=120°$	1 Bravais
Triclinic: $a\neq b\neq c$, $\alpha\neq\beta\neq\gamma$	1 Bravais (most general)

MILLER INDICES

(hkl) → single plane	Round brackets
{hkl} → family of planes	Curly brackets
[uvw] → direction	Square brackets
Steps: Intercept→Recip→LCM	Finding (hkl)

IMPORTANT VALUES

$d(100) : d(110) : d(111)$	Cubic d-ratio
$= 1 : 1/\sqrt{2} : 1/\sqrt{3}$	Values
$d_{min} = \lambda/2$	Min detectable d
Total Bravais = 14	Crystal systems = 7

CRYSTAL STRUCTURE
= Lattice + Basis

14 BRAVAIS
3+2+4+2+1+1+1

HIGHEST APF
FCC = 74%

BRAGG'S θ
Glancing angle (from plane)

UNIT II — ADDITIONAL CONCEPTS & NUMERICALS GUIDE

NUMERICAL SOLVING STRATEGIES

Type 1: Find d from Bragg's Law

1. Write: $n\lambda = 2d \sin\theta$
2. Rearrange: $d = n\lambda / (2 \sin\theta)$
3. Substitute values (ensure λ and d same units)

Type 2: Find lattice parameter 'a'

1. Find d from Bragg's law: $d = n\lambda/2\sin\theta$
2. Use cubic spacing: $a = d \times \sqrt{h^2+k^2+l^2}$
3. Identify the plane (hkl) from given information

4. Calculate. If 2θ given, halve it first.

4. Compute $\sqrt{h^2+k^2+l^2}$ and multiply

📊 Type 3: Find Miller Indices from intercepts

1. Given intercepts: say 2a, 3b, 6c → values: 2, 3, 6
2. Take reciprocals: 1/2, 1/3, 1/6
3. Multiply by LCM(2,3,6)=6: get 3, 2, 1
4. Answer: (321)

📊 Type 4: Calculate APF

1. Find number of atoms per cell (N): SC=1, BCC=2, FCC=4
2. Find radius in terms of a using diagonal relations
3. Volume of atoms = $N \times (4/3)\pi r^3$
4. APF = Volume of atoms / a^3

⭐ CONSTANTS & VALUES TO MEMORIZE

Quantity	Value	Context
X-ray wavelength (Cu Kα)	1.54 Å = 0.154 nm	Most common XRD source
1 Ångström	1 Å = 10^{-10} m = 0.1 nm	Unit conversion
APF (SC)	$\pi/6 \approx 0.5236$	Only 52% space utilized
APF (BCC)	$\sqrt{3}\pi/8 \approx 0.6802$	68% space utilized
APF (FCC)	$\pi/(3\sqrt{2}) \approx 0.7405$	Maximum for same-size spheres
√2	≈ 1.414	Face diagonal / a for cubic
√3	≈ 1.732	Body diagonal / a for cubic
Avogadro's Number	6.022×10^{23} mol^{-1}	Density calculations

DENSITY FORMULA (FROM CRYSTAL STRUCTURE)

$\rho = (N \times M) / (N_a \times a^3)$ where N = atoms/cell, M = molar mass, N_a = Avogadro's number, a = lattice parameter

🎯 ALL THE BEST FOR YOUR EXAM!

Remember: $n\lambda = 2d \sin\theta$ | $d = a/\sqrt{h^2+k^2+l^2}$ | FCC=74% | Crystal Structure = Lattice + Basis